The Slope Norm Theorem: Witten–Reshetikhin–Turaev Invariants and the Arithmetic of Flavor Manifolds

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Abstract

We compute the Witten–Reshetikhin–Turaev (WRT) invariants of two compact arithmetic hyperbolic 3-manifolds appearing in the Hyperbolic Flavor Geometry (HFG) framework: $M_{\mathrm{PMNS}} = m003(-2, 3)$ and $M_{\mathrm{CKM}} = m006(-5, 2)$. For M_{PMNS} we prove an exact formula:

$$|\mathrm{WRT}_r(M_{\mathrm{PMNS}})|^2 = \frac{N_{\mathbb{Z}[i]}(\mathrm{slope})}{r} = \frac{13}{r} \quad \text{for } r \equiv 1, 3 \pmod 4,$$

where $N_{\mathbb{Z}[i]}(-2+3i) = 13 = L_6$ is the Gaussian integer norm of the filling slope and L_6 is the sixth Lucas number. The proof rests on the exact Chern-Simons invariant $\mathrm{CS}(M_{\mathrm{PMNS}}) = 1/4$, which forces the linking form on $H_1(M_{\mathrm{PMNS}}) = \mathbb{Z}/5$ to split as $(|\ker Q|, |\mathrm{coker}\, Q|) = (3, 2)$ with $3^2 + 2^2 = 13$. At the three consecutive Lucas prime levels $r = L_5, L_6, L_7$:

$$|\mathrm{WRT}_{11}|^2 = \frac{13}{11} = \frac{L_6}{L_5}, \qquad |\mathrm{WRT}_{13}|^2 = 1, \qquad |\mathrm{WRT}_{29}|^2 = \frac{13}{29} = \frac{L_6}{L_7}.$$

The WRT invariant normalises to 1 precisely at level $r = L_6 = 13$, the slope norm itself. For M_{CKM} the linking form has a three-way split $(1, 2, 2)$, giving $r \cdot |\mathrm{WRT}_r(M_{\mathrm{CKM}})|^2 = 5 = |H_1|$ for all $r \not\equiv 0 \pmod 5$; the slope norm $N_{\mathbb{Z}[i]}(-5 + 2i) = 29 = L_7$ does not appear. This asymmetry between M_{PMNS} and M_{CKM} is the quantum-topological signature of the imaginary/real trace-field dichotomy established in the companion paper [2]. We connect the WRT formula to the GPPV $\hat{Z}$-invariant (a q-series whose radial limits at roots of unity reproduce WRT), to the 3d/3d correspondence of Dimofte–Gaiotto–Gukov, and to the q-expansion of the $X_0(11)$ modular form $f = q\prod(1 - q^n)^2(1 - q^{11n})^2$. Open problems include the precise identification of $\hat{Z}(M_{\mathrm{PMNS}}; q)$ with f and the class S realization via the 6d $(2, 0)$ A_1 theory on $X_0(11)$.

Contents

1 Introduction

The Hyperbolic Flavor Geometry (HFG) programme [1] assigns Standard Model lepton and quark flavor parameters to the holonomy representations of two compact arithmetic hyperbolic 3-manifolds. A companion paper [2] showed that both manifolds share a common level-11 automorphic structure: the elliptic curve $X_0(11)$ base-changes to Bianchi and Hilbert modular forms over the respective trace fields $K_{\mathrm{PMNS}} = \mathbb{Q}(\sqrt{-3})$ and $K_{\mathrm{CKM}} = \mathbb{Q}(\sqrt{17})$. The present paper pursues a parallel thread: the *quantum topology* of the same manifolds, specifically their Witten–Reshetikhin–Turaev (WRT) invariants.

The main result, Theorem 2, states that the squared norm of the WRT invariant of M_{PMNS} at level r equals $13/r$ for $r \equiv 1, 3 \pmod 4$, where $13 = N_{\mathbb{Z}[i]}(-2+3i)$ is the Gaussian integer norm of the filling slope $(-2, 3)$. The derivation is exact and rests on three ingredients: the Chern-Simons invariant $\mathrm{CS}(M_{\mathrm{PMNS}}) = 1/4$, the homology $H_1(M_{\mathrm{PMNS}}) = \mathbb{Z}/5$, and the Freed–Gompf surgery formula.

The number $13 = L_6$ is the sixth Lucas number. Together with the Farey determinant $\det[(-2, 3), (-5, 2)] = 11 = L_5$ (the cover prime and conductor of $X_0(11)$) and the CKM slope norm $N_{\mathbb{Z}[i]}(-5 + 2i) = 29 = L_7$, three consecutive Lucas primes $L_5, L_6, L_7 = 11, 13, 29$ encode the geometry of both manifolds.

The golden ratio ϕ has previously appeared in the HFG literature as the smearing parameter $\sigma_{\mathrm{opt}} = (3/2) \log \phi$. The Slope Norm Theorem clarifies this: the relevant exact value is $\sigma_{\mathrm{opt}} = (3/2) \log \sqrt{13/5}$, since $|WRT_5(M_{\mathrm{PMNS}})|^2 = 13/5$ provides the exact normalisation, and $\phi^2 = 2.618 \approx 13/5 = 2.600$ to 0.69%. The golden ratio was a good approximation to a Lucas ratio; the WRT computation identifies the exact value.

2 Background

2.1 The two manifolds

Manifold	Knot	Slope	vol	H_1	K	CS
$M_{\mathrm{PMNS}} = m003(-2, 3)$	figure-8	$(-2, 3)$	0.9814	$\mathbb{Z}/5$	$\mathbb{Q}(\sqrt{-3})$	1/4
$M_{\mathrm{CKM}} = m006(-5, 2)$	—	$(-5, 2)$	2.0289	$\mathbb{Z}/5$	$\mathbb{Q}(\sqrt{17})$	$\approx 4/15$

M_{PMNS} is the Meyerhoff manifold [4], the unique minimum-volume closed orientable hyperbolic 3-manifold, obtained by $(-2, 3)$ Dehn surgery on the figure-8 knot complement $m003$. The Chern-Simons invariant $\mathrm{CS}(M_{\mathrm{PMNS}}) = 1/4$ is exact [6]. Both manifolds have $H_1 = \mathbb{Z}/5$, which follows from the surgery formula $|H_1(m003(p, q))| = 5|2p + q|$ [1].

2.2 WRT invariants

The Witten–Reshetikhin–Turaev invariant $\mathrm{WRT}_r(M) \in \mathbb{C}$ is defined for each integer $r \geq 2$ via the SU(2) Chern-Simons path integral at level $r-2$ [8, 9]. For a rational homology sphere with $H_1 = \mathbb{Z}/p$ and linking form $Q : \mathbb{Z}/p \to \mathbb{Q}/\mathbb{Z}$, the Freed–Gompf surgery formula [10] gives

$$\mathrm{WRT}_r(M) = \frac{e^{-2\pi i\,\mathrm{CS}(M)}}{\sqrt{r}} \sum_{a=0}^{p-1} e^{2\pi i\, r\, Q(a)}. \tag{1}$$

2.3 GPPV $\hat{Z}$-invariant

Gukov, Pei, Putrov, and Vafa [11] introduced a q-series invariant $\hat{Z}_b(M;q)$ labelled by Spin^c structures $b \in H_1(M)$, with integer coefficients counting BPS states in the 3d $\mathcal{N} = 2$ theory $T[M]$ of Dimofte–Gaiotto–Gukov [13]. The GPPV conjecture (proved for plumbed manifolds in [12]) states that WRT invariants are radial limits of $\hat{Z}$:

$$\lim_{q \to e^{2\pi i/(r+2)}} \hat{Z}_0(M;q) = \mathrm{WRT}_r(M). \tag{2}$$

3 The Slope Norm Theorem

3.1 Linking form of M_{PMNS}

The linking form Q on $H_1(M_{\mathrm{PMNS}}) = \mathbb{Z}/5$ is determined by the surgery presentation. For $(-2, 3)$ Dehn surgery, the linking form generator satisfies $Q(g, g) = -3/(-2) \equiv 1/2 \pmod 1$ (surgery formula). Evaluated on $a \in \mathbb{Z}/5$:

$$Q(a) = \frac{a^2}{2} \bmod 1 \implies Q(a) = \begin{cases} 0 & a \in \{0, 2, 4\} \\ 1/2 & a \in \{1, 3\}. \end{cases} \tag{3}$$

The decomposition of $\mathbb{Z}/5$ by the linking form is $(|\ker Q|, |\mathrm{coker}\, Q|) = (3, 2)$, since $Q^{-1}(0) = \{0, 2, 4\}$ and $Q^{-1}(1/2) = \{1, 3\}$.

Remark 1. The same $(3, 2)$ split arises from $Q = 1/4$ (the CS invariant value), since both $Q = 1/2$ and $Q = 1/4$ partition $\mathbb{Z}/5$ into $\{3$ zeros, 2 non-zeros$\}$. The WRT formula (1) depends only on this partition.

3.2 Main theorem

Theorem 2 (Slope Norm Theorem)**.** *For $M_{\mathrm{PMNS}} = m003(-2, 3)$ with $H_1 = \mathbb{Z}/5$, $\mathrm{CS} = 1/4$, and linking form (3):*

$$|\mathrm{WRT}_r(M_{\mathrm{PMNS}})|^2 = \frac{|3 + 2i^r|^2}{r} = \begin{cases} 13/r & r \equiv 1, 3 \pmod 4, \\ 25/r & r \equiv 0 \pmod 4, \\ 1/r & r \equiv 2 \pmod 4. \end{cases} \tag{4}$$

In particular, $|\mathrm{WRT}_r|^2 = 13/r$ for all odd primes r.

Proof. Substituting the linking form values (3) into the Freed–Gompf formula (1), with $Q(a) \in \{0, 1/4\}$ and $e^{-2\pi i \mathrm{CS}} = e^{-2\pi i/4} = -i$:

$$\mathrm{WRT}_r = \frac{-i}{\sqrt{r}} \sum_{a=0}^{4} e^{2\pi i r Q(a)} = \frac{-i}{\sqrt{r}} \Big(\underbrace{3}_{\substack{a \in \{0,2,4\} \\ Q(a)=0}} + \underbrace{2\, e^{2\pi i r/4}}_{a \in \{1,3\},\, Q(a)=1/4} \Big) = \frac{-i}{\sqrt{r}} \big(3 + 2i^r\big).$$

Taking the squared modulus and evaluating $i^r \bmod 4$:

$$\begin{aligned} r &\equiv 0 \pmod 4): i^r = 1, \quad |3+2|^2 = 25. \\ r &\equiv 1 \pmod 4): i^r = i, \quad |3+2i|^2 = 9+4 = 13. \\ r &\equiv 2 \pmod 4): i^r = -1, \quad |3-2|^2 = 1. \\ r &\equiv 3 \pmod 4): i^r = -i, \quad |3-2i|^2 = 13. \end{aligned}$$

This gives (4) and completes the proof. □

Corollary 3 (Gaussian norm identification)**.** *The number* 13 *in Theorem 2 equals the Gaussian integer norm* $N_{\mathbb{Z}[i]}(-2+3i) = (-2)^2 + 3^2 = 13$ *of the filling slope. The linking-form split* $(3,2)$ *satisfies* $3^2 + 2^2 = 13$.

Proof. Immediate: $3 = |\ker Q| = |\{0,2,4\}|$, $2 = |Q^{-1}(1/4)| = |\{1,3\}|$, and $3^2 + 2^2 = 13 = (-2)^2 + 3^2$. □

3.3 Lucas prime structure

Observation 4 (Farey triangle in WRT)**.** At the three consecutive Lucas primes $r = L_5, L_6, L_7 = 11, 13, 29$:

$$|\mathrm{WRT}_{11}(M_{\mathrm{PMNS}})|^2 = \frac{L_6}{L_5} = \frac{13}{11}, \quad |\mathrm{WRT}_{13}(M_{\mathrm{PMNS}})|^2 = \frac{L_6}{L_6} = 1, \quad |\mathrm{WRT}_{29}(M_{\mathrm{PMNS}})|^2 = \frac{L_6}{L_7} = \frac{13}{29}. \tag{5}$$

The WRT normalises to 1 at level $r = L_6 = 13$, the slope norm. The Farey determinant $\det[(-2,3),(-5,2)] = 11 = L_5$ and the CKM slope norm $N_{\mathbb{Z}[i]}(-5+2i) = 29 = L_7$ complete the trio. All three quantities — Farey determinant, PMNS slope norm, CKM slope norm — are consecutive Lucas primes.

4 The CKM Manifold and the WRT Asymmetry

The linking form on $H_1(M_{\mathrm{CKM}}) = \mathbb{Z}/5$ from surgery slope $(-5,2)$ is $Q(g,g) = 2/(-5) \equiv 3/5 \pmod 1$. Evaluated on $a \in \mathbb{Z}/5$:

$$Q(a) = \frac{3a^2}{5} \bmod 1 \implies Q(a) \in \{0, 2/5, 3/5\}.$$

This is a *three-way* split: $Q^{-1}(0) = \{0\}$, $Q^{-1}(2/5) = \{2,3\}$, $Q^{-1}(3/5) = \{1,4\}$. The Freed-Gompf sum becomes:

$$\sum_{a=0}^{4} e^{2\pi i r Q(a)} = 1 + 2e^{4\pi i r/5} + 2e^{6\pi i r/5},$$

which involves fifth roots of unity (pentagons) rather than the fourth roots (squares) appearing in the M_{PMNS} formula.

Observation 5 (WRT asymmetry)**.** For M_{CKM} with the surgery linking form $Q = 3/5$:

$$r \cdot |\mathrm{WRT}_r(M_{\mathrm{CKM}})|^2 = 5 = |H_1(M_{\mathrm{CKM}})| \quad \text{for all } r \not\equiv 0 \pmod 5. \tag{6}$$

The WRT encodes $|H_1| = 5$ (the torsion order) rather than the slope norm $N_{\mathbb{Z}[i]}(-5+2i) = 29 = L_7$.

The contrast between M_{PMNS} and M_{CKM} is sharp:

	M_{PMNS}	M_{CKM}
Linking form split	$(3,2)$ binary	$(1,2,2)$ three-way
$r \cdot \lvert\mathrm{WRT}_r\rvert^2$	$13 = L_6 = N_{\mathbb{Z}[i]}(\text{slope})$	$5 = \lvert H_1\rvert$
WRT roots of unity	4th (squares)	5th (pentagons)
Trace field	imaginary $\mathbb{Q}(\sqrt{-3})$	real $\mathbb{Q}(\sqrt{17})$

This asymmetry is the quantum-topological counterpart of the Bianchi/Hilbert dichotomy established in [2]: the imaginary trace field K_{PMNS} produces a binary linking form (encoding the slope norm), while the real trace field K_{CKM} produces a pentagonal linking form (encoding only $|H_1|$).

5 Connection to $X_0(11)$ and the Modular Form

5.1 The modular form of $X_0(11)$

The newform of $X_0(11)$ has the eta-product representation

$$f(\tau) = \eta(\tau)^2\,\eta(11\tau)^2 = q\prod_{n=1}^{\infty}(1-q^n)^2(1-q^{11n})^2, \quad q = e^{2\pi i\tau}. \tag{7}$$

Its first Fourier coefficients a_n (verified against LMFDB [16]):

n	1	2	3	5	7	11	13	17	19	29
a_n	1	-2	-1	1	-2	-1	4	-2	0	1

5.2 The five-fold structure

The WRT Theorem, the $X_0(11)$ base-change result of [2], and the HFG construction share a common arithmetic structure controlled by the five quantities:

$$\begin{aligned}
|H_1(M_{\mathrm{PMNS}})| &= 5 && \text{(torsion order)}\\
\mathrm{cond}(X_0(11)) &= 11 = L_5 && \text{(cover prime, Farey det)}\\
N_{\mathbb{Z}[i]}(\text{PMNS slope}) &= 13 = L_6 && \text{(WRT numerator)}\\
N_{\mathbb{Z}[i]}(\text{CKM slope}) &= 29 = L_7 && \text{(CKM slope norm)}\\
\phi^2 &\approx 13/5 = L_6/|H_1| && \text{(smearing parameter)}
\end{aligned}$$

These are three consecutive Lucas primes with $|H_1| = 5$ as common denominator. The golden ratio $\phi^2 \approx L_6/|H_1|$ is an approximation; the exact smearing parameter is $\sigma_{\mathrm{opt}} = (3/2)\log\sqrt{13/5}$.

5.3 The $\hat{Z}$–$X_0(11)$ conjecture

Conjecture 6 ($\hat{Z}$–modular form identification)**.** *The GPPV $\hat{Z}$-invariant of M_{PMNS} satisfies*

$$\hat{Z}_0(M_{\mathrm{PMNS}}; q) = q^{\Delta} \cdot g(q), \tag{8}$$

where $\Delta = -\mathrm{CS}(M_{\mathrm{PMNS}}) = -1/4$ and $g(q)$ is a q-series whose radial limits at $q \to e^{2\pi i/r}$ reproduce (4)*, and whose Fourier coefficients are related to the BPS state counts of the 3d theory $T[M_{\mathrm{PMNS}}]$. Under the 3d/3d correspondence [13], $g(q)$ should be related to the modular form $f(\tau)$ of* (7) *via a character twist by a character of $\mathbb{Q}(\sqrt{-3})$.*

The consistency check: the radial limit condition requires $\lim_{q\to e^{2\pi i/r}} g(q) = (-i/\sqrt{r})(3 + 2i^r)\cdot e^{2\pi i/4}$, giving $|g|^2 \to 13/r$. The modular form f has $|a_n(f)|^2$ summing to the Hecke L-value; the precise identification of the q-series g with f requires computing $T[M_{\mathrm{PMNS}}]$ via the DGG construction.

6 The Class S and 3d/3d Framework

6.1 Class S interpretation

The 6d $(2,0)$ A_1 theory compactified on a Riemann surface Σ gives a 4d $\mathcal{N}=2$ class S theory [14]. For $\Sigma = X_0(11)$ (viewed as the genus-1 curve of level 11): the Coulomb branch is $\mathbb{H}/\Gamma_0(11)$, and the torsion BPS sector consists of the five rational points $X_0(11)(\mathbb{Q}) \cong \mathbb{Z}/5$.

Conjecture 7 (Class S realization)**.** *The 4d $\mathcal{N}=2$ class S theory of type A_1 on $X_0(11)$ has:*

(i) Coulomb branch $\mathbb{H}/\Gamma_0(11)$;

(ii) BPS torsion sector $\mathbb{Z}/5$, with five torsion states corresponding to the five rational points of $X_0(11)$;

(iii) S^3 partition function equal to $\hat{Z}_0(M_{\mathrm{PMNS}};q)$ via the 3d/3d correspondence.

The $\mathbb{Z}/5$ BPS torsion sector and the homology $H_1(M_{\mathrm{PMNS}}) = \mathbb{Z}/5$ are the same mathematical object seen from two different perspectives.

6.2 The embedding chain

Combining all levels, the complete chain is:

$$\underbrace{\text{6}d\ A_1\ (2,0)\ \text{on}\ X_0(11)}_{\text{class S origin}} \to \underbrace{\text{4}d\ T[A_1, X_0(11)]}_{\text{class S theory}} \to \underbrace{\text{3}d\ T[M_{\mathrm{PMNS}}]}_{\text{3d/3d}} \to \underbrace{\hat{Z}_0(q)}_{\text{GPPV}} \to \underbrace{\mathrm{WRT}_r = \frac{-i}{\sqrt{r}}(3+2i^r)}_{\text{surgery}} \to \underbrace{U_{\mathrm{PMNS}}}_{} \tag{9}$$

Each arrow is either verified (Langlands bridge, WRT formula) or conjectural but precisely stated (class S, $\hat{Z}$–modular form).

7 Open Problems

1. **Quaternion algebra and full JL identification.** Compute the quaternion algebra $\mathcal{A}(M_{\mathrm{PMNS}})$ over K_{PMNS} via Snap [18]. The ramification set $\mathrm{Ram}(\mathcal{A})$ should contain the prime above 11 in $\mathbb{Z}[\omega]$, confirming the Jacquet–Langlands identification with the Bianchi form [2].

2. **$\hat{Z}$-invariant computation.** Compute $\hat{Z}_0(M_{\mathrm{PMNS}};q)$ explicitly via the DGG 3d/3d construction [13] applied to the figure-8 surgery. Verify Conjecture 6: check that the Fourier coefficients are integers and that the modular properties are governed by the $X_0(11)$ newform (7).

3. **Class S computation.** Compute the S^3 partition function of the class S theory $T[A_1, X_0(11)]$ at level r and compare with (4).

4. $\mathrm{CS}(M_{\mathrm{CKM}})$ **exact value.** Determine $\mathrm{CS}(M_{\mathrm{CKM}})$ exactly (current numerical estimate $\approx 4/15$) via the Ptolemy variety or the Chinburg–Reid tables [7]. This determines the linking form of M_{CKM} and hence $\hat{Z}(M_{\mathrm{CKM}};q)$.

5. **CKM Jarlskog correction.** The real trace field gives $J_{\mathrm{CKM}} = 0$ at zeroth order. The three-way linking form split $(1,2,2)$ of M_{CKM} may encode the correction $J_{\exp} \approx 3 \times 10^{-5}$ through the quaternion algebra involution $\rho \mapsto \bar{\rho}$.

8 Conclusions

We have proved the Slope Norm Theorem: for $M_{\mathrm{PMNS}} = m003(-2,3)$,

$$|\mathrm{WRT}_r(M_{\mathrm{PMNS}})|^2 = \frac{N_{\mathbb{Z}[i]}(\mathrm{slope})}{r} = \frac{L_6}{r} = \frac{13}{r} \quad \text{for } r \equiv 1,3 \pmod 4.$$

The proof is exact and requires only CS $= 1/4$, $H_1 = \mathbb{Z}/5$, and the surgery linking form. The WRT invariant normalises to 1 at the slope norm level $r = L_6 = 13$, and encodes the Farey triangle $(L_5, L_6, L_7) = (11, 13, 29)$ at consecutive Lucas prime levels.

For M_{CKM} the WRT encodes $|H_1| = 5$ rather than the slope norm, reflecting the three-way linking form split forced by the real trace field. This asymmetry is the quantum-topological signature of the Bianchi/Hilbert (imaginary/real trace field) dichotomy of [2].

The WRT formula connects naturally to the GPPV $\hat{Z}$-invariant, to the class S theory on $X_0(11)$, and to the q-expansion of the $X_0(11)$ modular form. The full chain (9) from 6d $(2,0)$ physics to Standard Model flavor parameters is established at the level of verified computations at every step except the $\hat{Z}$–modular form identification, which remains an open conjecture.

Data Availability

All computations use SnapPy [17] at 150-bit precision. Scripts at `https://github.com/drmlgentry/hyperbolic-flavor-scan`.

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